



ITE/ASSOCIATION

refers to the topographic position. Association is occurrence of certain feature in relation with other. Certain features are not directly identifiable by appearance in an image but could be interpreted easily according to its relationship with the surroundings.

relationship of feature to the surrounding features provides clues toward identity.

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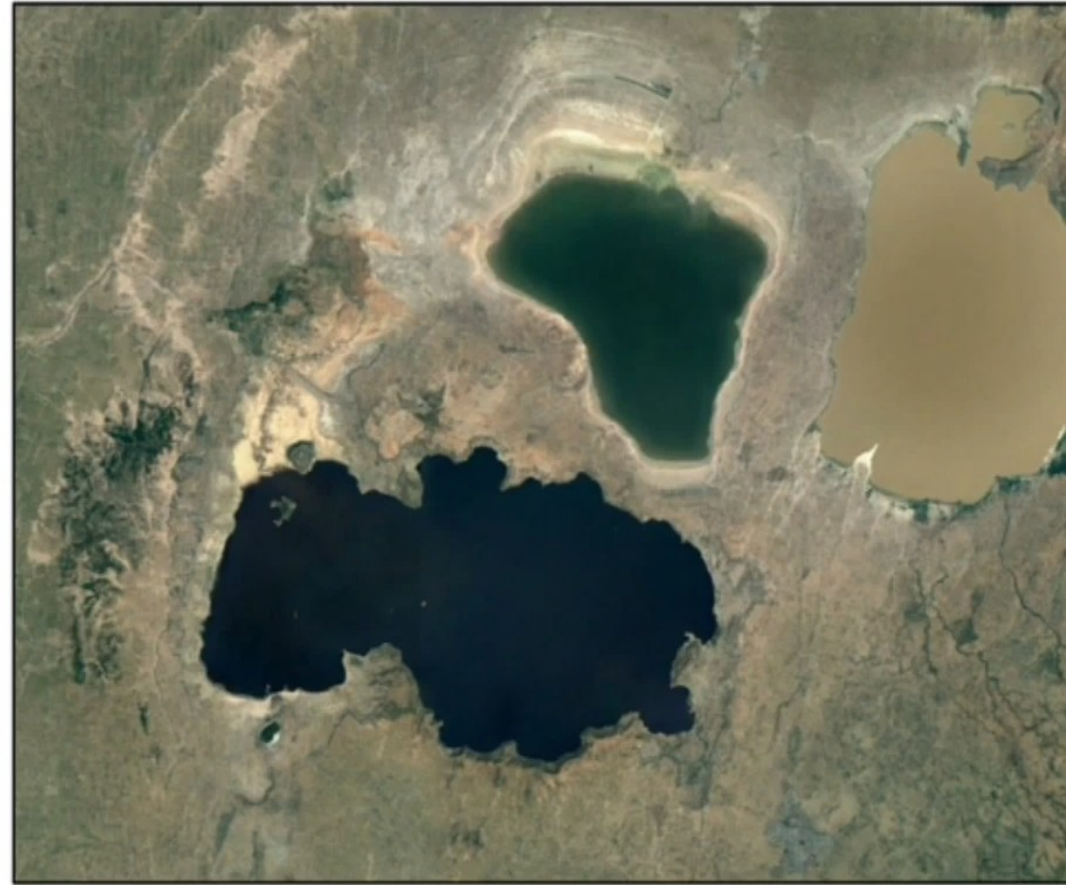






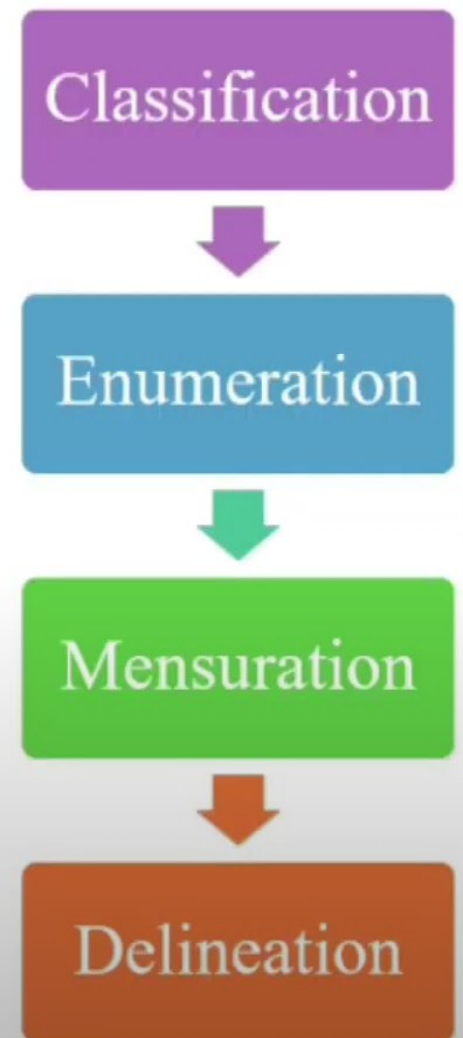
SUAL IMAGE TERPRETATION

OTE SENSING & GIS

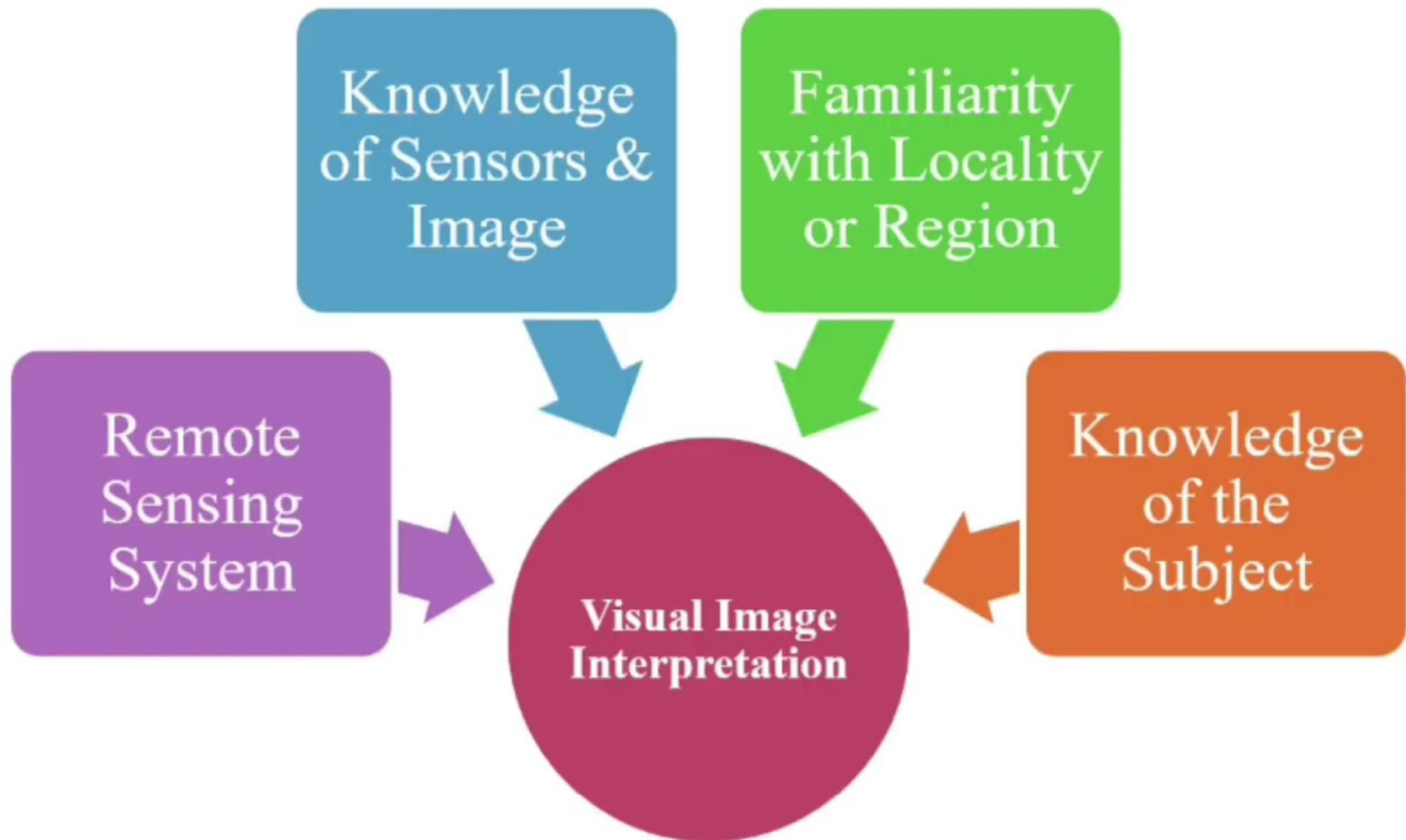


INTRODUCTION

Visual image interpretation is a process of identifying features seen on the images by an analyst/interpreter and communication of information obtained from these images to others for evaluating their significance. This process, however, is not restricted to making decisions concerning what objects appear in images; it also includes determination of their relative locations and extents.

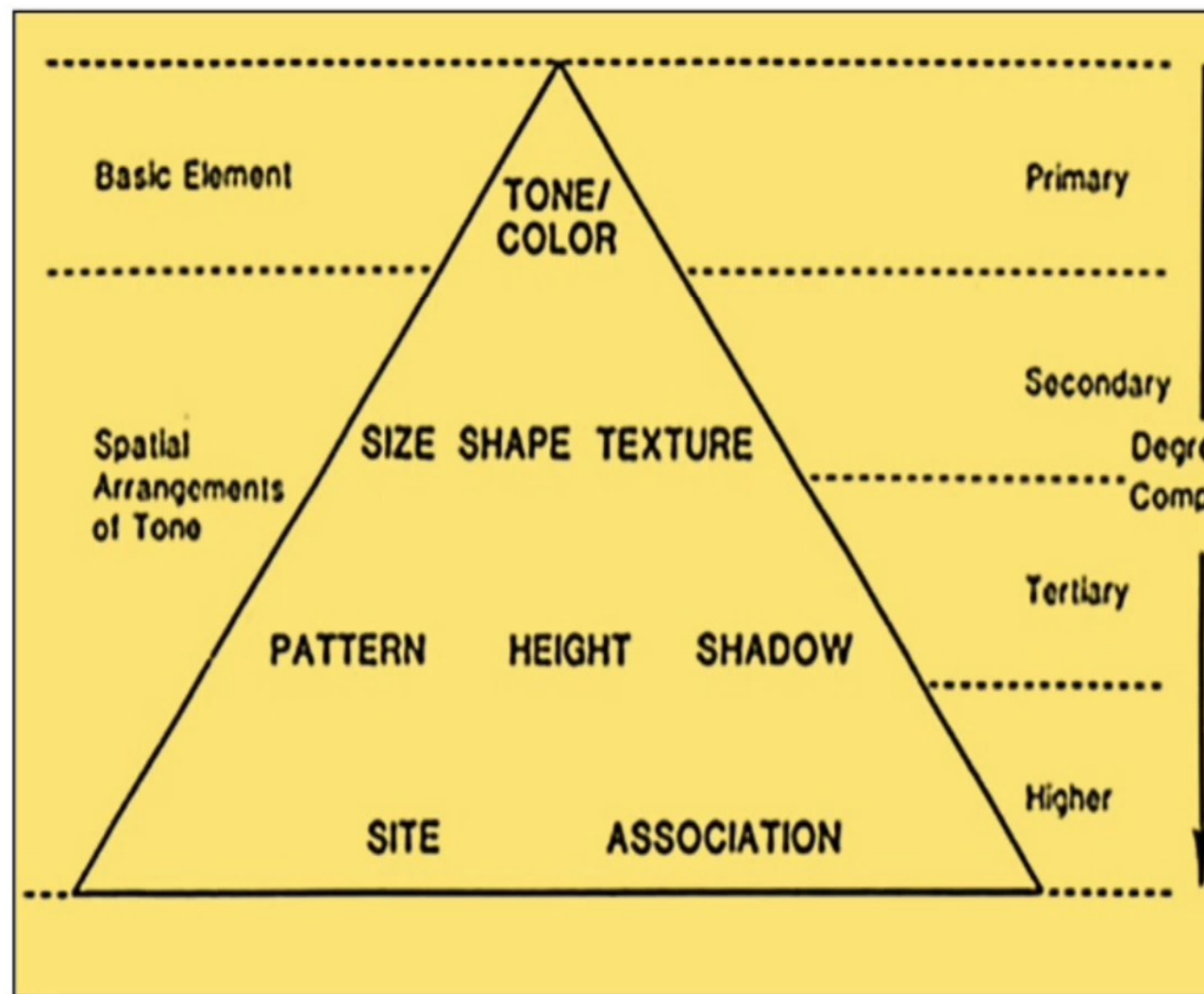


REQUIREMENTS FOR VISUAL IMAGE INTERPRETATION



ELEMENTS OF VISUAL IMAGE INTERPRETATION

ase of interpretation of
llite images, these
racteristics of objects
studied with reference
single or multiple
pectral bands because
e are generally more
n one images acquired
ifferent spectral
ons of electromagnetic
pectrum



ONE/COLOR

dictionary definition of Tone is the particular quality of brightness, deepness, or intensity of a shade of a color.

It refers to relative brightness or color of a feature on an image. The tonal variation makes it easier to differentiate between various features on an image.

Tonal images is influenced by the following factors:

- reflectivity of the object

- intensity of reflected light

- geographic latitude

- transmission of filters and

- photographic processing.





SIZE

Size of the Feature is an important feature in distinguishing between Similar Features.

Objects in an image is a function of scale hence, the objects must be considered in the context of the size of a graph/image



SHAPE

Shape relates to the general form, configuration or outline of an individual object. Shape is one of the most important single factors for recognizing objects from images.

Regular geometric shapes are usually indicators of human presence and activity. Similarly, irregular shapes are usually indicators of natural objects.

Some objects can be identified almost solely on the basis of their shapes, for example Railway Lines, Sports Stadiums, Air Strips etc.



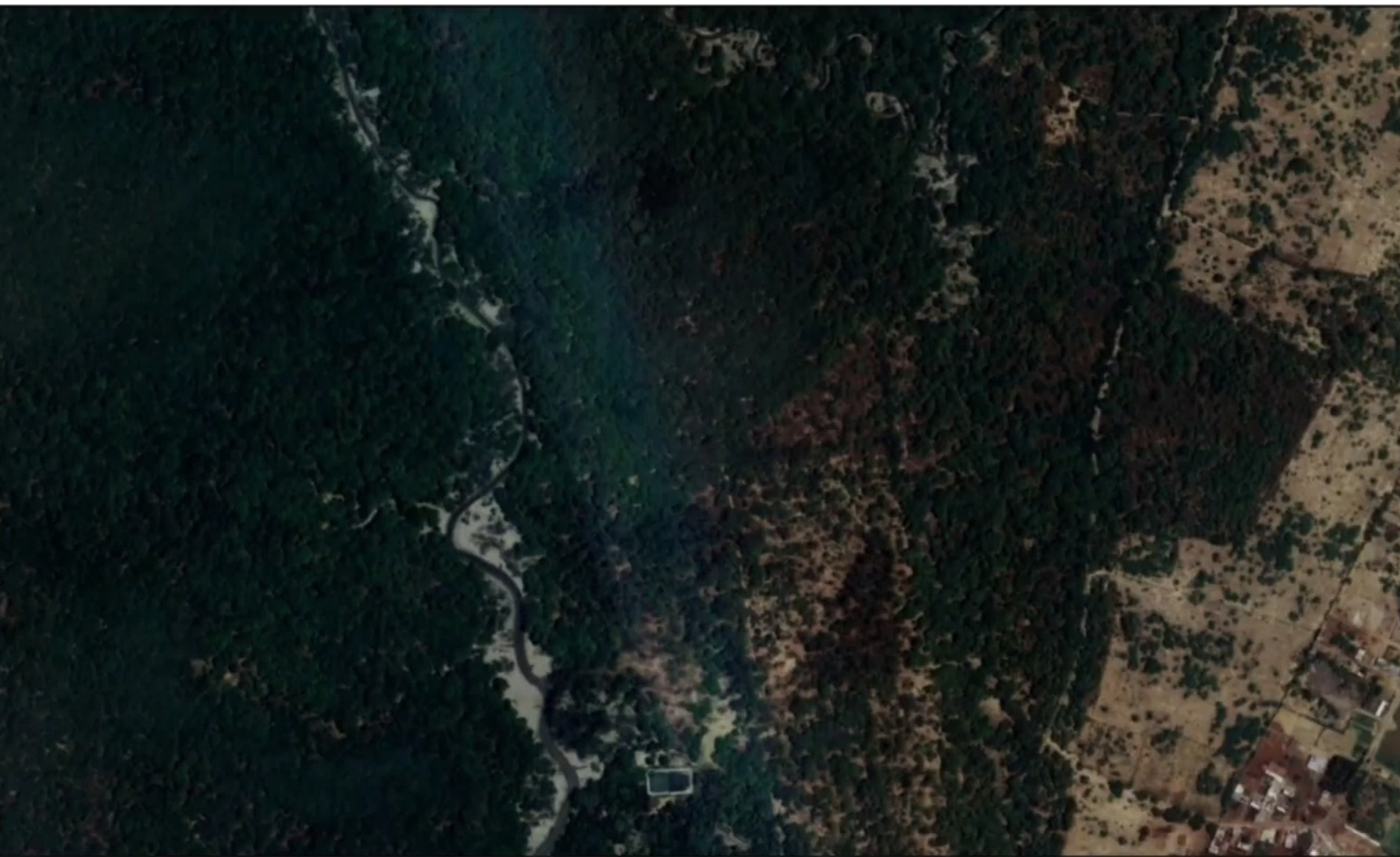
TEXTURE

Texture is an expression of roughness or smoothness as exhibited by the images. It is the rate of change of tonal values (frequency of tonal changes).

Texture signifies the frequency of change and arrangement of tones in an image and is produced by an aggregate of unit features too small to be clearly recognized individually on an image.

Texture may be related to Surface Roughness, a property that is, in particular, important in Microwave region of EMS.

Texture can be expressed qualitatively as coarse, moderate, fine, very fine, smooth, rough, rippled and mottled.



PATTERN

Pattern develops in an image due to spatial arrangement of objects. Hence, pattern can be defined as the spatial arrangement of objects in an image. Certain objects can be easily identified because of their pattern. A particular pattern may have its genetic relation with several factors of its origin.

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For example, urban and rural settlement areas can be easily identified based on the patterns created by the rows of houses or buildings.



HADOW

Shadow refers to a dark area produced by the features coming between the sun rays and surface. Shadows in remote sensing images have both advantages as well as disadvantages at the same time.

One advantage is it provides certain relative height information of the features and also an idea of the terrain profile which is an aid to interpretation while the features under shadows are not very well identifiable is a disadvantage. So due consideration must be given to shadows while interpreting an image.